Kwanele Maziya

LinkedIn: www.linkedin.com/in/knmaziya
GitHub: github.com/Kwanele08
Email: neleziya08@gmail.com
Mobile: +91-8544902763

SKILLS

- **Languages**: Python, C, C++, Java
- **Frameworks**: Streamlit, Scikit-learn
- **Tools/Platforms**: Streamlit, Automation Anywhere, Google Firebase, One Signal, MixPanel, MySQL, MongoDB
- **Soft Skills**: Communication, Adaptability, Leadership, Problem Solving

INTERNSHIPS

Ummo Inc. September 2022 - January 2023

- *User Engagement Intern*
  - Analyzed user behavior and engagement patterns on the Ummo app using MixPanel event data and insights.
  - Identified key interaction trends to inform targeted communication strategies and improve user retention.
  - Designed and deployed both in-app and off-app notifications through OneSignal to enhance engagement and trigger meaningful user actions.
  - **Tech stacks used**: One Signal, MixPanel, Google Firebase

Innovators Association of Eswatini January 2023 - March 2023

- *User Engagement Intern*
  - Collaborated with teammates in creating a digital self-directed search test for high school leaving students preparing for tertiary education.
  - My duty, user engagement monitoring; I worked on MongoDB to incorporate explanatory suggestions to simplify jargon so that students could read with better understanding.
  - **Tech stacks used**: MongoDB, MixPanel

PROJECTS

- **Deep Learning-Based Steganography**: October 2025

- Built an image-in-image hiding pipeline using a convolutional encoder-decoder architecture.
- Implemented a 2-bit LSB steganography baseline for comparison.
- Built Added Gaussian noise during training to improve reconstruction robustness.
- Evaluated model performance using PSNR, SSIM, and Mean Squared Error.
- Ran reproducible experiments across 100 fixed image pairs.
- Generated comparison graphs and CSV-based experimental results.
- Repository Link: https://github.com.kwanele08/stego
- **Tech Stacks:** Python, PyTorch, OpenCV, CNN

- **Digital Reverse Logistics For Damaged Goods**: March – May 2025

- Developed an ML-enabled platform using Streamlit to optimize reverse logistics processes for damaged goods.
- Built a Random Forest classifier to predict damaged returns and provide data-driven insights into return patterns.
- Implemented clustering algorithms to identify trends in return behaviour and optimize logistics decisions.
- Designed an interactive dashboard enabling users to upload data, visualize clusters, and make predictions in real time.
- Integrated synthetic transport optimization and data analysis features to support efficient reverse supply chain management.
- Repository Link: https://github.com/kwanele08/reverese-logistics
- **Tech Stacks**: Python, Streamlit.

TRAINING

- **Mastering C++ :**
  Lovely Professional University June – July 2025
- Completed structured training focused on C++ fundamentals, Object-Oriented Programming (OOP), and Data Structures & Algorithms (DSA).
- Gained hands-on experience with core OOP concepts including encapsulation, abstraction, class design, and modular programming.
- Implemented key data structures such as circular queues, stacks, and arrays, and explored their application in real-world problem solving.
- Strengthened algorithmic reasoning, debugging techniques, and the ability to write efficient, optimized C++ code.
- Enhanced logical thinking and confidence in translating theoretical concepts into working solutions through practical assignments and mini-implementations.

CERTIFICATES

- C++ And Object-Oriented Programming – Lovely Professional University Link — January 2024
- Lean Six Sigma Yellow Belt – Alison Link — November 2023

ACHIEVEMENTS

- **International Robotics Competition Participation**: October 2022
  Selected to represent my country in an Olympic style robotics competition as a coach (FIRST Global Challenge 2022 Geneva).
- **Entrepreneurship Competition**:

  Part of the UNESWA Kwaluseni Enactus team that obtained third place in the national entrepreneurship competition called Enactus.

EDUCATION

- **Lovely Professional University** — Punjab, India
  Bachelor of Technology - Computer Science and Engineering; ***CGPA: 6.9*** — *Since August 2023*
- **University Of Eswatini** — Matsapha, Eswatini
  Bachelor of Science – Information Science — *2020 – 2022(Paused)*
- **Lusoti High School** — Lubombo, Eswatini
  Intermediate; ***Percentage: 70%*** — *2014 - 2018*